

ANNUAL EQUIPMENT INSPECTION REPORT

Fiscal Year 2025 -- Petromax Baytown Complex

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Reviewed by: K. Tanaka, Plant Reliability Manager

1. Executive Summary

This report presents the findings from the Annual Equipment Inspection program conducted across Units 100 through 500 at the Petromax Baytown Refining Complex during the period January 6, 2025 through March 10, 2025. A total of 47 equipment items were inspected as part of the API 510/570/653 inspection program. The inspection scope included pressure vessels, heat exchangers, rotating equipment, storage tanks, and associated piping systems.

Overall, 38 equipment items (81%) were found in satisfactory condition with no immediate action required. Six items (13%) require corrective actions within 90 days, and three items (6%) have been flagged for priority attention. Most notably, Centrifugal Feed Pump P-101 exhibited significant mechanical seal degradation and bearing wear that warrants immediate seal replacement before the Q2 turnaround. Heat Exchanger E-301 showed tube bundle fouling exceeding 40% of design capacity, requiring chemical cleaning during the next planned outage.

2. Inspection Summary by Unit

Tag	Description	Type	Condition	Next Due
P-101	Centrifugal Feed Pump	API 610 Pump	PRIORITY	Q2 2025
P-102	Booster Pump	API 610 Pump	SATISFACTORY	Q1 2026
V-201	Flash Separator	Pressure Vessel	MONITOR	Q3 2025
E-301	Shell & Tube HX	Heat Exchanger	PRIORITY	Q2 2025
C-401	Recip. Compressor	API 618 Compressor	SATISFACTORY	Q1 2026
TK-501	Crude Storage Tank	API 653 Tank	MONITOR	Q4 2025

3. Detailed Findings -- Unit 100 (Feed System)

3.1 P-101 -- Centrifugal Feed Pump

Equipment Tag: P-101

Manufacturer: Flowserve (Model: HPX 6x8x13)

Installed: August 2018

Service: Crude oil feed to atmospheric distillation column

Operating Conditions: 285 degF, 165 psig discharge, 3,200 GPM rated

Inspection Findings:

The mechanical seal (John Crane Type 4620) was found with visible leakage at the outboard seal face. Seal leakage rate measured at 12 drops/minute, exceeding the 3 drops/minute threshold per API 682. Vibration analysis conducted on January 22, 2025 revealed elevated readings at the drive-end bearing: 0.42 in/s peak velocity at 1x running speed (3,560 RPM), compared to the baseline of 0.18 in/s. Spectrum analysis indicates possible shaft misalignment. The coupling (Rexnord Omega E40) showed wear marks on the elastomeric element consistent with angular misalignment of approximately 0.008 inches.

Recommendations:

1. Replace mechanical seal assembly (John Crane Type 4620) -- Priority: HIGH
 2. Perform laser alignment of pump-motor coupling during seal replacement
 3. Replace coupling elastomeric element (Rexnord P/N E40-FLEX)
 4. Re-establish vibration baseline after corrective maintenance
 5. Schedule follow-up vibration survey within 30 days of repair
- Target completion: Before April 15, 2025 (Q2 turnaround window)

3.2 P-102 -- Booster Pump

Equipment Tag: P-102

Manufacturer: Sulzer (Model: MSD-D 150/6)

Installed: August 2018

Service: Intermediate pressure boost, downstream of P-101

Operating Conditions: 310 degF, 320 psig discharge, 2,800 GPM rated

Inspection Findings:

P-102 was found in satisfactory operating condition. Vibration levels at all measurement points are within acceptable limits per ISO 10816-3 (Zone A/B boundary). Bearing temperatures recorded at 168 degF drive-end and 155 degF

non-drive-end, both within normal operating range. Mechanical seal (Flowserve QBW) showing no visible leakage. Suction strainer differential pressure reading 2.1 psi, below the 5 psi alarm threshold. Lubrication system oil analysis (Shell Morlina S4 B 220) returned satisfactory results -- water content < 100 ppm, particle count ISO 16/13/10.

Recommendations:

1. Continue routine condition monitoring per existing PM schedule
2. Next comprehensive inspection due Q1 2026

4. Detailed Findings -- Unit 200 (Separation)

4.1 V-201 -- Flash Separator Vessel

Equipment Tag: V-201

Manufacturer: Boardman LLC

Design Code: ASME Section VIII, Div. 1

Installed: June 2016

Service: Two-phase flash separation of crude feed

Operating Conditions: 380 degF, 95 psig, two-phase crude/gas

Inspection Findings:

Ultrasonic thickness (UT) measurements taken at 24 CML locations per API 510 inspection plan. Minimum measured wall thickness: 0.487 inches at CML-12 (lower shell course, 6 o'clock position), compared to nominal 0.625 inches. Calculated corrosion rate: 6.9 mils/year (long-term average). Remaining life at current rate: approximately 14 years, which exceeds the next inspection interval. Internal inspection via robotic crawler revealed minor pitting corrosion (maximum pit depth 0.045 inches) on the bottom head near the boot drain nozzle. Level instrumentation (Rosemount 5300 guided wave radar) verified and calibrated -- deviation within 0.5% of span.

Recommendations:

1. Continue monitoring CML-12 annually via UT spot check
2. Evaluate internal coating application at next turnaround (2027)
3. Next full API 510 internal inspection due Q3 2025 (aligned with planned outage)

5. Detailed Findings -- Unit 300 (Heat Exchange)

5.1 E-301 -- Shell and Tube Heat Exchanger

Equipment Tag: E-301

Manufacturer: Alfa Laval (TEMA Type: BEM)

Installed: March 2017

Service: Crude preheat via hot product return stream

Design: 450 tubes, 3/4" OD x 16 BWG, 20 ft effective length

Shell Side: Hot product, 420 degF inlet / 290 degF outlet

Tube Side: Crude feed, 250 degF inlet / 370 degF outlet

Design Pressure: 300 psig (shell) / 250 psig (tube)

Inspection Findings:

Performance monitoring data indicates heat transfer coefficient has declined by 38% over the past 12 months. Shell-side pressure drop has increased from 4.2 psi to 11.7 psi, indicating significant fouling. Eddy current testing (ECT) of the tube bundle performed on February 14, 2025 identified 12 tubes with wall loss exceeding 40% -- these tubes have been plugged. An additional 23 tubes showed wall loss between 20-40%. Total plugged tube count is now 31 out of 450 (6.9%), which is within the 10% maximum allowable per engineering evaluation. Tube sheet ligament measurements at the tube-to-tubesheet joints showed no significant erosion.

Recommendations:

1. Schedule chemical cleaning of shell side during Q2 2025 turnaround -- Priority: HIGH
2. Consider tube bundle replacement if plugged tube count exceeds 45 (10%) at next inspection
3. Install shell-side fouling monitor (DP transmitter) for real-time trending
4. Evaluate antifouling coating (Curran International Plastacor) for tube bundle OD
5. Increase tube-side velocity by 15% to reduce fouling tendency (requires P-101 capacity review)

6. Detailed Findings -- Unit 400 (Gas Compression)

6.1 C-401 -- Reciprocating Compressor

Equipment Tag: C-401

Manufacturer: Ariel Corporation (Model: JGK/4)

Installed: November 2019

Service: Off-gas compression from flash separator V-201 to fuel gas header

Operating Conditions: 1st stage 15 psig suction / 65 psig discharge; 2nd stage 60 psig suction / 185 psig discharge

Driver: 500 HP electric motor, 1,200 RPM

Inspection Findings:

All four cylinder bores measured within acceptable wear limits. Piston rod runout: 0.0015 inches TIR (tolerance 0.003 inches). Valve inspection revealed minor wear on 2nd stage discharge valve plates -- valve lift reduced to 0.085 inches from nominal 0.100 inches. Packing leakage rates within API 618 limits at 2.8 SCFM. Crosshead pin bearings inspected and found satisfactory. Rod load calculations verified within 95% of rated capacity. Pulsation dampener (Peerless orifice plate type) condition satisfactory.

Recommendations:

1. Plan 2nd stage discharge valve replacement at next opportunity (non-urgent, within 6 months)
2. Continue packing leakage monitoring -- replacement recommended if exceeding 5 SCFM
3. Next comprehensive API 618 inspection due Q1 2026

7. Detailed Findings -- Unit 500 (Storage)

7.1 TK-501 -- Crude Oil Storage Tank

Equipment Tag: TK-501

Manufacturer: CB&I (now McDermott)

Design Code: API 650

Installed: 2012

Service: Crude oil intermediate storage

Capacity: 250,000 barrels

Dimensions: 200 ft diameter x 48 ft shell height

Roof Type: External floating roof with double-deck pontoon

Inspection Findings (API 653 External):

Shell plate thickness measurements taken at 36 CML locations. Minimum shell thickness: 0.372 inches at Course 1, CML-7 (nominal 0.500 inches). Long-term corrosion rate on Course 1: 5.2 mils/year (soil-side). Floating roof condition: primary seal (mechanical shoe) in good condition, secondary seal (wiper type) showing wear on approximately 30% of circumference. Roof drain system functional -- no blockages observed. Settlement survey indicates uniform settlement of 1.2 inches over 13 years of service, within API 653 limits. Foundation ring wall shows no cracking or displacement. Cathodic protection system verified -- tank-to-soil potential -0.92V CSE (criterion: more negative than -0.85V CSE).

Recommendations:

1. Replace secondary seal during next opportunity (within 12 months) -- Priority: MODERATE
2. Continue annual external inspection per API 653
3. Internal inspection (out-of-service) recommended by Q2 2027
4. Verify cathodic protection rectifier output quarterly

8. Appendix -- Inspection Personnel & Certifications

The following certified inspection personnel participated in this annual program:

R. Vasquez -- API 510, API 570, API 653, ASNT Level II (UT, MT, PT, ECT)
J. Morrison -- API 510, API 570, ASNT Level II (UT, RT)
L. Chen -- API 653, CWI (Certified Welding Inspector)
M. Okonkwo -- Vibration Analyst Category III (ISO 18436-2)
S. Patel -- API 618 (Reciprocating Compressors), Machinery Diagnostic Specialist

This report has been prepared in accordance with API Recommended Practice 580 (Risk-Based Inspection) and the Petromax Refining Corp. Mechanical Integrity Program Manual (PM-MI-001, Rev. 8). All findings and recommendations are subject to review by the Plant Reliability Manager and the Process Safety Management Coordinator.